

POLARIS DC-1 — TECHNICAL SPECIFICATION

SPEC-04: FINISHES AND DATA HALL CONTAINMENT

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Project: Polaris DC-1 — Aviapolis Data Campus, Phase 1, Vantaa, Finland
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Synthetic demonstration data. All parties, values and figures in this document are fictional.

1. Scope

- 1.1 This specification covers internal finishes to the white space, electrical/mechanical support and battery rooms, the Administration/NOC block and the gatehouse, and the cold-aisle containment framework and blanking panels within the four data halls (1,250 m² white space per hall, 5,000 m² total).
- 1.2 **Interface boundary.** The containment framework and blanking panels are Contractor's scope. IT racks, servers, network equipment and containment **doors** are excluded — these are supplied and installed by the Tenant, per PDC-ER-001 §1.3 and §9. The Contractor shall coordinate framework dimensions, door-leaf openings and locking-hardware provisions with the Tenant's containment door supplier but shall not procure or install the doors themselves.
- 1.3 This specification excludes the structural frame and precast walls (PDC-SPEC-02) and the external envelope (PDC-SPEC-03), to which the finishes described here are applied.

2. Standards and References

Reference	Subject
IEC 61340-4-1 / SFS-EN 1081	Electrical resistance of floor coverings — static-dissipative flooring
SFS-EN 13501-1	Reaction to fire classification of construction products, internal finishes
National Building Code of Finland	Internal finish fire-classification requirements
SFS-EN 50600-2-1	Information technology — data centre facilities and infrastructures: building construction
SFS-EN 1090-1, SFS-EN 1090-2	Execution of steel structures (containment framework, where metal)
SFS-EN 13501-2	Fire classification — resistance to fire (blanking panel assemblies at compartment lines)
SFS-EN 1307	Textile floor coverings — classification (Admin/NOC carpet tile)

3. Materials

3.1 White space (5,000 m² total, 1,250 m² per hall)

Element	Specification
Floor	Sealed, static-dissipative coating applied directly to the structural ground-bearing slab (PDC-SPEC-01) — no raised access floor
Floor electrical resistance	1×10^6 to $1 \times 10^9 \Omega$, per IEC 61340-4-1, verified before handover
Walls	Painted precast concrete, sealed joints, light-reflective finish
Ceiling / soffit	Exposed structure, painted, coordinated with overhead containment within the 2.4 m service zone below the 6.6 m underside-of-steel level

3.2 Electrical/mechanical support and battery rooms

Element	Specification
Floor	Sealed concrete; chemical-resistant coating in battery rooms, resistant to electrolyte spillage
Walls	Painted precast/blockwork
Ceiling / soffit	Exposed structure
Battery room segregation	Physical segregation between UPS/battery modules per PDC-BOD-001 §5.2, coordinated with the fire-rated construction of each battery room

3.3 Administration/NOC block and gatehouse

Area	Floor	Walls	Ceiling
Admin offices	Carpet tile, SFS-EN 1307 class suited to commercial use	Plasterboard, painted	Suspended acoustic tile
NOC	Raised access floor (cabling only — this is the sole raised-floor area on the project) with carpet tile finish	Plasterboard, painted	Suspended acoustic tile
Gatehouse	Sealed concrete / vinyl	Painted blockwork	Suspended tile

3.4 Containment framework and blanking

Element	Specification
Containment framework (cold-aisle)	Powder-coated steel or aluminium framing system, coordinated to the rack layout grid provided by the Tenant
Blanking panels	Rigid, fire-rated where crossing a compartment line, sized to close all rack-position gaps not occupied by Tenant equipment at handover
Containment doors	Excluded — Tenant supply. Framework openings sized and hardware-prepared to receive the Tenant's doors
Roof/ceiling panels over containment (if part of the system)	Fire-rated to match the containment system's tested configuration

4. Workmanship and Execution

4.1 Floor coating. The white-space floor coating shall be applied to a slab surface prepared and tested for moisture content per the coating manufacturer's requirements, in a minimum of two coats, achieving uniform coverage and the electrical resistance range in Section 3.1.

4.2 Containment coordination. Containment framework shall be erected only after the overhead service zone (busway, cable tray, pipework — outside this specification's scope) has been coordinated at shop-drawing stage, so that no clash occurs within the 2.4 m service zone beneath the 4.2 m minimum clear height.

4.3 Interface demarcation. The Contractor shall issue an interface coordination drawing for

each hall clearly identifying the boundary between the Contractor's containment framework and blanking scope and the Tenant's rack, server, network equipment and containment-door scope, countersigned by the Tenant's representative before framework installation begins in that hall.

4.4 Battery room finishes. Chemical-resistant coatings in battery rooms shall be installed after all cable containment and electrical first-fix in that room is complete, to avoid damage to the finish by follow-on trades.

4.5 Sequencing with Tenant fit-out. In Hall 1, where Tenant fit-out access begins at Month 12 in parallel with Section 1 completion (PDC-ER-001 §9), the Contractor shall complete and hand over the containment framework in discrete, inspectable stages so that Tenant door installation can follow without obstructing the Contractor's remaining Section 1 completion works.

5. Tolerances

Item	Tolerance
Floor coating thickness	As specified by the manufacturer, minimum $\pm 10\%$
Floor coating flatness	Consistent with the underlying FM2 slab (PDC-SPEC-01) — no additional levelling compound permitted that would alter flatness class
Containment framework alignment to hot/cold aisle centreline	± 10 mm
Containment framework level (top rail)	± 5 mm
Blanking panel fit at rack gaps	No visible air gap exceeding 3 mm
Painted wall surface flatness	5 mm over 2 m straightedge

6. Testing and Inspection

Test / inspection	Frequency	Acceptance basis
Floor electrical resistance test	Minimum 5 locations per hall, post-cure	Within 1×10^6 – $1 \times 10^9 \Omega$ range, IEC 61340-4-1
Coating adhesion/thickness test	Sample locations per hall	Meets manufacturer's specification
Fire classification verification of finishes	Document review, all painted/coated systems	Certificates match the classification required by the National Building Code
Battery room chemical-resistance verification	Visual and manufacturer sign-off, each battery room	No evidence of coating failure under test exposure
Containment framework structural/load inspection	100% of installed framework, before blanking	No deflection or connection defects against the system's design
Interface coordination drawing sign-off	Before framework installation in each hall	Countersigned by Tenant's representative

7. Submittals

Submission	Timing
Finishes schedule and material samples (floor coating, paint, carpet tile)	S2 detailed design
Static-dissipative floor coating test certificates	Before floor coating order is placed
Containment framework shop drawings, coordinated with overhead M&E containment	S3 construction issue

Interface coordination drawing (Contractor/Tenant boundary, per hall)	Before framework installation, countersigned by the Tenant
Battery room chemical-resistance coating data sheet	Before application
O&M and cleaning/maintenance instructions for all finishes and containment systems	Per Section handover

8. Spare Parts and Warranties

8.1 The Contractor shall hand over spare blanking panels (minimum 5% of the installed quantity per hall) and a floor-coating touch-up kit sufficient for minor repairs, at each Section's O&M handover.

8.2 Floor coatings and containment framework shall carry the manufacturer's standard product warranty; the Contractor's workmanship warranty period for finishes and containment installation shall be a minimum of 12 months from the relevant Section's Taking-Over, without prejudice to any longer statutory liability period under Finnish law.

End of PDC-SPEC-04 Rev A.

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